



RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform

Predictive Demand • Customer Segmentation • Churn Analysis
• Inventory Optimization

Advanced Data Science & Analytics Project Zidio Development / Portfolio / Interview

Reference March 2026



Title: RetailPulse – AI-Powered Customer Analytics & Demand Forecasting Platform

subtitle: End-to-End Data Science & Analytics Solution for Retail Demand Prediction & Customer Insights

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Prepared-for: Zidio Development – Data Science & Analytics Domain

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version: 2.0 – Industry Edition

"Retailers lose billions due to poor demand forecasting and stock mismanagement. RetailPulse uses advanced analytics and machine learning to predict demand, segment customers, detect churn, and optimize inventory – helping retailers reduce stockouts by 30-50% and increase revenue by 15-25%."



1. Business Case & Production Objectives

Mission

Build an end-to-end data science platform that ingests sales, customer, and inventory data to deliver accurate demand forecasts, customer segmentation, churn prediction, and inventory optimization recommendations for retail businesses.

Why This Project Matters for Zidio Development

Retail clients need data-driven decisions to reduce waste and maximize profit. RetailPulse provides a complete analytics solution that Zidio can offer to supermarket chains, fashion retailers, and e-commerce companies.

Quantified Business Impact Targets

- Reduce stockouts by **30-50%** through accurate demand forecasting
- Increase revenue by **15-25%** through better inventory decisions
- Improve customer retention by identifying at-risk customers early
- Process **10M+ transactions** per month with daily batch jobs under 5 minutes

Non-Functional Requirements

- Model accuracy: MAPE $\leq$ 12% for demand forecasting
- Processing time: < 5 minutes for daily batch jobs
- Scalability: Handle 10M+ transactions per month
- Observability: Full MLflow tracking and drift detection

2. Core Functional Requirements

ID	Capability	Detailed Description & Business Value	Key Acceptance Criteria & Production Metrics
F-01	Data Ingestion & Cleaning	Ingest sales, customer, and inventory data from multiple sources	Automated ETL pipeline, data quality checks
F-02	Customer Segmentation	RFM + behavioral segmentation using K-Means / DBSCAN	6-8 meaningful segments with business interpretation
F-03	Demand Forecasting	Time-series forecasting with Prophet + LSTM ensemble	MAPE $\leq$ 12%, 30-day ahead predictions
F-04	Churn Prediction	Classification model to identify at-risk customers	AUC-ROC $\geq$ 0.88, precision@top 20% $\geq$ 0.75
F-05	Inventory Optimization	Recommend reorder quantities using forecasted demand	Reduce overstock/understock by 25-40%
F-06	Interactive Analytics Dashboard	Streamlit dashboard with visualizations and what-if analysis	Real-time insights, exportable reports

3. Production Technology Stack – 2026

Layer	Primary Technology	Rationale / Alternatives
Language	Python 3.11	Data science ecosystem
Data Processing	Pandas, NumPy, Scikit-learn	Core data manipulation and ML
Forecasting	Prophet + LSTM (PyTorch)	Hybrid time-series forecasting
Dashboard	Streamlit	Fast interactive analytics
Experiment Tracking	MLflow	Model versioning and reproducibility
Database	PostgreSQL + Redis	Structured data + caching
Containerization	Docker	Consistent deployment
Orchestration	Kubernetes	Scalable production deployment

Layer	Primary Technology	Rationale / Alternatives
Monitoring	Prometheus + Grafana + Evidently AI	Drift detection and performance monitoring

4. 28-Days Day-by-Day Execution Plan

Week 1 – Data Exploration & Preparation

Day 1

- Dataset selection (retail sales, customer, inventory data)
- Initial EDA notebook: distribution analysis, missing values, correlation heatmap

Day 2

- Data cleaning and feature engineering (RFM scores, rolling statistics)
- Data validation with Great Expectations

Day 3

- Customer segmentation using K-Means and DBSCAN
- Cluster evaluation and business interpretation

Day 4

- Time-series data preparation for forecasting
- Stationarity tests and decomposition

Day 5

- Baseline Prophet model for demand forecasting

Day 6

- LSTM model implementation with PyTorch Lightning

Day 7

- Week 1 checkpoint: EDA report, cleaned dataset, baseline models logged in MLflow

Week 2 – Advanced Modeling & Churn Prediction

Day 8

- Hybrid forecasting model (Prophet + LSTM ensemble)

Day 9

- Churn prediction model using XGBoost with SHAP explainability

Day 10

- Inventory optimization logic using forecasted demand

Day 11

- Feature importance analysis and model tuning with Optuna

Day 12

- Drift detection setup using Evidently AI

Day 13

- Automated retraining pipeline with Airflow

Day 14

- Week 2 checkpoint: Forecasting and churn models ready, optimization logic implemented

Week 3 – Dashboard & Analytics Layer

Day 15

- Streamlit dashboard skeleton with multi-page layout

Day 16

- Demand forecasting visualizations and what-if analysis

Day 17

- Customer segmentation and churn risk dashboard

Day 18

- Inventory optimization recommendations UI

Day 19

- Real-time metrics and alerts

Day 20

- Export functionality (CSV/PDF reports)

Day 21

- Week 3 checkpoint: Fully interactive dashboard with all insights

Week 4 – Deployment & Production Polish

Day 22

- Docker multi-stage builds for the application

Day 23

- Kubernetes manifests and deployment configuration

Day 24

- GitHub Actions CI/CD pipeline

Day 25

- Cloud deployment on AWS or GCP

Day 26

- Monitoring setup with Prometheus and Grafana

Day 27

- Load testing and final accuracy validation

Day 28

- Final QA, README polishing, demo video recording, PDF export

5. Challenges, Learnings & Industry Best Practices

- Handling non-stationary time-series data with proper decomposition
- Balancing model accuracy with interpretability using SHAP
- Best practices: MLflow for reproducibility, Evidently AI for drift detection, Airflow for orchestration

6. Security & Privacy Highlights

- Data anonymization for customer records
- Role-based access in the dashboard
- Secure API endpoints with JWT
- Audit logging for sensitive operations

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Project Submission Guidelines

March 2026 Edition

Prepared for participants

Focus: Industry-grade data science, ML, and analytics projects

Submission period: You need to submit the project on or before the due date. If you submit after the due date, you will be automatically disqualified for stipend by the system.

Evaluation emphasis: Model quality · Documentation · Live demo · Business impact & MLOps awareness

1. General Rules & Eligibility

- All code and models must be **original work** created mainly during the ZIDIO preparation/submission window.
- Participants must submit **exactly 1 project** (as per the planned structure).
- Projects should demonstrate **progressive complexity**:
 1. Strong data exploration & feature engineering
 2. Advanced modeling (forecasting, segmentation, prediction)
 3. MLOps / production readiness (drift detection, retraining, deployment)
- Plagiarism / AI-generated content will result in disqualification for stipend.

2. Mandatory Submission Deliverables

Submit **one consolidated package** containing the project:

#	Deliverable	Format / Location	Required?	Evaluation Weight
1	Project Documentation / Report (1 PDF)	1 PDF file (A4, 10–18 pages)	Yes	25%
2	Live Public Demo URL	1 HTTPS link (Streamlit / Hugging Face / AWS)	Yes	30%
3	GitHub Repository	1 repo with clear folders & notebooks	Yes	20%
4	README.md	Detailed, professional README in repo	Yes	15%
5	Demo Video(s)	4–8 min (YouTube unlisted / Loom / Drive)	Yes	10%

Naming convention recommendation

RetailPulse – AI-Powered Customer Analytics & Demand
Forecasting_Zidio_March2026.pdf/zip

3. Documentation Guidelines – What the PDF should contain

Use this consistent structure for a polished, professional portfolio feel.

1. Hero / Cover Section

- Large project title
- Catchy tagline
- Your name + date

- Gradient background (optional but recommended)

2. Project Overview

- Vision & objectives
- Target users / use cases
- Business value delivered
- Non-functional goals (accuracy, latency, scalability targets)

3. Key Features (table format)

| ID | Feature | Description | Acceptance Criteria |

4. Technology Stack (table)

| Category | Technology | Rationale / Alternatives |

5. Architecture Overview

- High-level data flow and MLOps pipeline
- Include screenshot (Excalidraw / Draw.io) or clean ASCII art

6. Detailed Execution Timeline

- Day-by-day or week-by-week breakdown
- Major deliverables per phase
- Milestones & checkpoints

7. Technical Highlights

- Model performance metrics (MAPE, AUC, F1)
- Feature engineering details
- MLOps practices (drift detection, retraining, MLflow)
- Challenges faced & how solved

8. Deployment & Operations

- Platform used (AWS, GCP, Streamlit Cloud, etc.)
- CI/CD pipeline summary
- Monitoring / drift detection setup

9. Visuals

- 8–15 high-quality screenshots / GIFs (EDA plots, dashboards, model outputs)
- Architecture diagram
- Live demo captures

10. Personal Reflection (strongly recommended)

- Key learnings
- Industry best practices applied
- Future roadmap ideas

4. Code & Repository Expectations

- Clean, modular notebooks and scripts
- Consistent naming & formatting (black / ruff)
- Semantic commit messages (`feat: add LSTM forecasting model` , `fix: drift detection threshold`)
- Feature branches & pull requests (even if solo)
- `.gitignore` properly configured (no large models or secrets)
- No committed secrets / API keys
- Basic tests / validation scripts appreciated
- Reproducible environment (requirements.txt or environment.yml)

5. Live Demo Requirements

- Publicly accessible (no VPN / geo-restriction)
- HTTPS enforced
- Core functionality available without sign-up (demo data OK)
- Fast initial load (< 8 seconds preferred)
- Mobile responsive where applicable
- Include brief instructions on demo page if non-obvious

6. Evaluation Criteria (Suggested 100-point scale)

Category	Points	Focus Areas
Innovation & Problem Solving	15	Original features, thoughtful business impact
Technical Depth & Model Quality	25	Feature engineering, model selection, evaluation rigor
MLOps & Production Readiness	20	Drift detection, retraining, deployment, monitoring
Documentation Quality	20	Clear, professional, well-structured
Deployment & Reliability	10	Stable live demo, easy local setup
Presentation & Polish	10	Demo video quality, visual appeal of docs

7. Important Rules & Tips

- **Strict deadline** – no late submissions accepted
- **File size limits:** ZIP $\leq$ 500 MB, individual PDFs $\leq$ 12 MB
- **No external dependencies that require payment / private keys** for judges to run demo
- **Backup plan:** If live demo is down, high-quality video + screenshots must still convey full functionality
- **Stand-out practices:**
 - Include model performance tables, SHAP plots, drift reports
 - Show real drift simulation in video
 - Mention MLOps maturity (MLflow, Evidently AI, retraining loop)
 - End docs with personal growth reflection

Submission note

Submit the project in the Zidio dashboard itself (no email submission).

Good luck!

Submit with confidence – your detailed planning and advanced topics already make these projects stand out.

Team Zidio Development

March 2026



Crafted with precision and modern data science principles • Zidio Development • March 2026

